

Top Students Tutoring Platform

Final Project Documentation

1. Project Overview

Project Idea

Top Students Tutoring Platform is a web-based educational platform that connects students who need academic support with verified top-performing students, also known as student tutors or batch toppers. The system allows students to search for tutors based on subjects, view tutor profiles, book online tutoring sessions, join meeting links, make payments, submit reviews, and send complaints when needed.

Unlike traditional tutoring platforms that depend only on professional teachers, this platform focuses on high-achieving students who have already studied the same subjects and can explain them in a simple and practical way. This makes the learning experience closer to the student's academic environment.

Purpose of the System

The purpose of the system is to make academic support easier, more organized, and more trustworthy for students. Many students face difficulties finding reliable tutors, comparing prices, checking availability, and booking sessions. At the same time, top students may have strong academic skills but do not have an organized platform to offer tutoring services.

This platform solves these problems by providing one system for tutor verification, search, booking, payment tracking, online meeting access, reviews, complaints, and admin management.

Main Features

The main features of the platform include:

- User registration and login.
- Role-based system for Students, Tutors, and Admin.
- Tutor profile creation.
- Tutor verification by Admin.
- Subject management.
- Tutor-subject relationship.
- Educational packages offered by tutors.
- Tutor availability management.
- Student booking system.
- Mock payment tracking.
- Platform commission calculation.

- Tutor earnings calculation.
- Online meeting link for each booking.
- Review and rating system.
- Complaint management system.
- Admin dashboard.
- Student dashboard.
- Tutor dashboard.
- Responsive front-end design using HTML, CSS, JavaScript, and Bootstrap.

Target Users

The system targets the following users:

Students

Students who need academic help in specific subjects, exam preparation, assignments, or course revision. They can search for tutors, view tutor profiles, book sessions, pay, join meetings, and review tutors.

Tutors / Top Students

High-achieving students who want to provide tutoring services. They can create profiles, add academic information, set hourly rates, choose subjects, create packages, manage availability, view bookings, and track earnings.

Admin

The admin is responsible for managing the system. The admin can approve or reject tutor profiles, manage users, view bookings, manage complaints, view reviews, and monitor platform activity.

2. Functional Requirements

Authentication and Authorization

1. The system shall allow users to register and log in.
2. The system shall support three main roles: Student, Tutor, and Admin.
3. The system shall allow each role to access only the pages and features related to that role.
4. The system shall prevent Students and Tutors from accessing Admin pages.
5. The system shall allow only approved Tutors to appear in student search results.
6. The system shall allow Admin to manage user roles and tutor verification status.

Student Functional Requirements

1. The student shall be able to create an account.
2. The student shall be able to log in and log out.
3. The student shall be able to view approved tutors.
4. The student shall be able to search for tutors by name or subject.
5. The student shall be able to view tutor profile details.
6. The student shall be able to view tutor academic information such as university, major, GPA, bio, hourly rate, and subjects.
7. The student shall be able to view educational packages offered by tutors.
8. The student shall be able to book a tutoring session.
9. The student shall be able to select subject, date, start time, and end time for a booking.
10. The student shall be able to view personal bookings.
11. The student shall be able to access the online meeting link for a confirmed booking.
12. The student shall be able to submit a review after a session.
13. The student shall be able to rate the tutor from 1 to 5.
14. The student shall be able to submit a complaint related to a booking or system issue.
15. The student shall be able to view booking status and payment status.

Tutor Functional Requirements

1. The tutor shall be able to create an account.
2. The tutor shall be able to log in and log out.
3. The tutor shall be able to create a tutor profile.
4. The tutor shall be able to add academic information such as university, major, GPA, bio, subjects, and hourly rate.
5. The tutor profile shall be pending until approved by Admin.
6. The tutor shall be able to update profile information.
7. The tutor shall be able to add available tutoring times.
8. The tutor shall be able to manage weekly availability.
9. The tutor shall be able to create educational packages.
10. The tutor shall be able to view received bookings.
11. The tutor shall be able to view student reviews.
12. The tutor shall be able to view payment information related to completed bookings.
13. The tutor shall be able to view basic statistics such as number of bookings, average rating, and total earnings.

Admin Functional Requirements

1. The admin shall be able to log in.
2. The admin shall be able to access the Admin Dashboard.
3. The admin shall be able to view all registered users.
4. The admin shall be able to view pending tutor profiles.
5. The admin shall be able to approve tutor verification requests.
6. The admin shall be able to reject tutor verification requests.
7. The admin shall be able to view all bookings.
8. The admin shall be able to view all payments.
9. The admin shall be able to view platform commission from payments.
10. The admin shall be able to view all reviews.
11. The admin shall be able to view all complaints.
12. The admin shall be able to update complaint status.
13. The admin shall be able to manage subjects.
14. The admin shall be able to monitor platform activity through dashboard statistics.

Booking Functional Requirements

1. The system shall allow students to create bookings with approved tutors.
2. Each booking shall include student, tutor, subject, booking date, start time, end time, status, payment status, and meeting link.
3. The system shall generate or store a meeting link for the booking.
4. The system shall display bookings in the Student Dashboard.
5. The system shall display bookings in the Tutor Dashboard.
6. The system shall prevent unapproved tutors from receiving bookings.
7. The system shall allow booking status tracking such as Confirmed, Pending, Completed, or Cancelled.

Payment Functional Requirements

1. The system shall store payment information for each booking.
2. Each payment shall be linked to a booking, student, and tutor profile.
3. The system shall store the total amount paid.
4. The system shall calculate platform commission.
5. The system shall calculate tutor earnings.
6. The system shall store payment method.

7. The system shall store payment status such as Pending or Paid.
8. The system may use mock payment instead of real payment gateway integration for the current version.

Review Functional Requirements

1. The system shall allow students to review tutors after sessions.
2. Each review shall include student, tutor, booking, rating, comment, and creation date.
3. The system shall display tutor reviews on the tutor profile page.
4. The system shall allow the admin to view reviews.

Complaint Functional Requirements

1. The system shall allow students or tutors to submit complaints.
2. Each complaint shall include submitted by user, booking if available, title, description, status, and creation date.
3. The admin shall be able to view complaints.
4. The admin shall be able to update complaint status such as Pending, In Review, Resolved, or Rejected.

CRUD Operations

The system includes the following CRUD operations:

Tutor Profiles

- Create tutor profile.
- Read tutor profile.
- Update tutor profile.
- Admin can update verification status.

Subjects

- Admin can create subjects.
- Users can view subjects.
- Admin can update subjects.
- Admin can delete subjects if needed.

Tutor Availability

- Tutor can create availability records.
- Tutor can view availability records.
- Tutor can update availability records.
- Tutor can disable or delete availability records.

Educational Packages

- Tutor can create packages.
- Students can view packages.
- Tutor can update packages.
- Tutor can delete packages.

Bookings

- Student can create bookings.
- Student, Tutor, and Admin can view bookings.
- Admin or system can update booking status.
- Booking records can be cancelled if needed.

Reviews

- Student can create reviews.
- Users can view reviews.
- Admin can manage reviews if needed.

Complaints

- Student or Tutor can create complaints.
- Admin can view complaints.
- Admin can update complaint status.

3. Non-Functional Requirements

Performance

1. The system should load pages quickly and efficiently.
2. Tutor search results should be displayed without unnecessary delay.
3. Database queries should be optimized using Entity Framework.
4. The system should avoid loading unnecessary data when displaying tutor lists and bookings.

Security

1. The system should use ASP.NET Identity for authentication.
2. Passwords should be securely hashed by ASP.NET Identity.
3. The system should use role-based authorization.
4. Admin pages should be protected from unauthorized access.

5. Student and Tutor dashboards should be accessible only by authenticated users.
6. The system should prevent users from accessing or modifying data that does not belong to them.
7. Database connection information should be stored securely in the application settings.

Usability

1. The system should have a simple and clear user interface.
2. Navigation should be easy for Students, Tutors, and Admin.
3. Forms should be clear and easy to understand.
4. Users should receive clear messages after actions such as booking, profile creation, payment confirmation, or complaint submission.
5. The booking process should be simple and understandable.

Scalability

1. The system should be designed to support more students, tutors, subjects, bookings, and payments in the future.
2. The database design should allow adding future features such as wallet system, coupon system, notifications, and real payment gateway.
3. The system should use separate tables for core entities to make future development easier.

Responsiveness

1. The front-end should be responsive on desktop, tablet, and mobile screens.
2. Bootstrap should be used to create a responsive layout.
3. Pages should maintain readability and usability on different screen sizes.
4. Forms, cards, tables, and navigation bars should adapt to different devices.

Reliability

1. The system should store user data, tutor profiles, bookings, payments, reviews, and complaints correctly.
2. The system should maintain data relationships using primary and foreign keys.
3. The database should be created using Entity Framework Code First and Migrations.
4. The system should handle basic errors and prevent invalid data submission.
5. The system should keep booking records even if tutor availability changes later.

4. Database Design and Connection

Database Technology

The project uses:

- SQL Server as the database management system.
- Entity Framework Core.
- Code First Approach.
- Migrations.
- ASP.NET Core Identity for user authentication and roles.

Main Database Tables

The database contains Identity tables and project-specific tables.

Identity Tables

AspNetUsers

Stores system users such as Students, Tutors, and Admins.

Main fields:

- Id: Primary Key
- FullName
- Address
- ProfileImageUrl
- UserName
- Email
- PasswordHash
- PhoneNumber
- CreatedAt

AspNetRoles

Stores system roles.

Main roles:

- Student
- Tutor
- Admin

AspNetUserRoles

Stores the relationship between users and roles.

Project Tables

TutorProfiles

Stores academic and tutoring information for top student tutors.

Fields:

- Id: Primary Key
- UserId: Foreign Key references AspNetUsers
- University
- Major
- GPA
- Bio
- Subjects
- HourlyRate
- VerificationStatus
- CreatedAt

Relationship:

- One user can have one tutor profile.
 - One tutor profile can have many bookings.
 - One tutor profile can have many reviews.
 - One tutor profile can have many educational packages.
 - One tutor profile can have many availability records.
 - One tutor profile can have many payments.
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Subjects

Stores subjects or courses available in the platform.

Fields:

- Id: Primary Key
- Name
- Description

Relationship:

- One subject can be linked to many tutor profiles through TutorSubjects.
 - One subject can be linked to many bookings.
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TutorSubjects

Stores the many-to-many relationship between TutorProfiles and Subjects.

Fields:

- Id: Primary Key
- TutorProfileId: Foreign Key references TutorProfiles
- SubjectId: Foreign Key references Subjects

Relationship:

- One tutor can teach many subjects.
 - One subject can be taught by many tutors.
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EducationalPackages

Stores ready educational packages created by tutors.

Fields:

- Id: Primary Key
- TutorProfileId: Foreign Key references TutorProfiles
- Title
- Description
- NumberOfHours
- Price
- CreatedAt

Relationship:

- One tutor profile can have many educational packages.

Examples of packages:

- Final Exam Review Package
 - Full Course Support Package
 - Assignment Help Package
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TutorAvailabilities

Stores tutor weekly availability.

Fields:

- Id: Primary Key

- TutorProfileId: Foreign Key references TutorProfiles
- DayOfWeek
- StartTime
- EndTime
- IsActive
- CreatedAt

Relationship:

- One tutor profile can have many availability records.

Purpose:

This table allows tutors to manage their available teaching times. Tutors can update availability when their schedule changes.

Bookings

Stores tutoring session bookings.

Fields:

- Id: Primary Key
- StudentId: Foreign Key referencesAspNetUsers
- TutorProfileId: Foreign Key references TutorProfiles
- SubjectId: Foreign Key references Subjects
- BookingDate
- StartTime
- EndTime
- Status
- PaymentStatus
- MeetingLink
- CreatedAt

Relationship:

- One student can create many bookings.
- One tutor profile can receive many bookings.
- One subject can be used in many bookings.
- One booking can have one payment.

- One booking can have one review.
 - One booking can be linked to complaints.
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Payments

Stores payment details for bookings.

Fields:

- Id: Primary Key
- BookingId: Foreign Key references Bookings
- StudentId: Foreign Key referencesAspNetUsers
- TutorProfileId: Foreign Key references TutorProfiles
- Amount
- PlatformCommission
- TutorEarning
- PaymentMethod
- PaymentStatus
- CreatedAt

Relationship:

- One booking can have one payment.
- One student can make many payments.
- One tutor profile can receive many payments.

Purpose:

This table supports the platform revenue model by storing the total amount, platform commission, and tutor earnings.

Reviews

Stores student reviews for tutors.

Fields:

- Id: Primary Key
- StudentId: Foreign Key referencesAspNetUsers
- TutorProfileId: Foreign Key references TutorProfiles
- BookingId: Foreign Key references Bookings

- Rating
- Comment
- CreatedAt

Relationship:

- One student can write many reviews.
- One tutor profile can receive many reviews.
- One booking can have one review.

Complaints

Stores complaints submitted by students or tutors.

Fields:

- Id: Primary Key
- SubmittedById: Foreign Key referencesAspNetUsers
- BookingId: Foreign Key referencesBookings, nullable
- Title
- Description
- Status
- CreatedAt

Relationship:

- One user can submit many complaints.
- One booking can be linked to many complaints.

Purpose:

This table supports the complaints and protection system for both students and tutors.

Database Relationships Summary

- AspNetUsers to TutorProfiles: One-to-One
- AspNetUsers to Bookings: One-to-Many as Student
- TutorProfiles to Bookings: One-to-Many
- Subjects to Bookings: One-to-Many
- TutorProfiles to Subjects: Many-to-Many through TutorSubjects
- TutorProfiles to EducationalPackages: One-to-Many

- TutorProfiles to TutorAvailabilities: One-to-Many
- Bookings to Payments: One-to-One
- Bookings to Reviews: One-to-One
- TutorProfiles to Reviews: One-to-Many
- AspNetUsers to Reviews: One-to-Many as Student
- AspNetUsers to Complaints: One-to-Many as SubmittedBy
- Bookings to Complaints: One-to-Many

5. Front-End Design

The front-end of the project is built using:

- HTML
- CSS
- JavaScript
- Bootstrap

The design includes all system roles:

- Student
- Tutor
- Admin

Public Pages

Home Page

The Home page introduces the platform and explains how students can connect with verified top student tutors.

About Page

The About page explains the purpose of the platform and the problem it solves.

Tutors List Page

Displays approved tutors with basic information such as name, major, subjects, hourly rate, and rating.

Tutor Details Page

Displays full tutor profile information including academic details, bio, subjects, packages, availability, and reviews.

Login Page

Allows registered users to log in.

Register Page

Allows new users to create accounts.

Student Pages

Student Dashboard

Displays student bookings, upcoming sessions, and quick links.

Search Tutors Page

Allows students to search and filter tutors by subject, name, price, or rating.

Book Session Page

Allows students to select subject, date, start time, and end time.

My Bookings Page

Displays all bookings made by the student.

Join Meeting Page

Allows students to open the online meeting link.

Add Review Page

Allows students to submit a rating and comment after a session.

Submit Complaint Page

Allows students to submit complaints related to a session or issue.

Tutor Pages

Tutor Dashboard

Displays tutor bookings, basic statistics, reviews, and earnings.

Create/Edit Tutor Profile Page

Allows tutors to create or update academic and tutoring information.

Manage Availability Page

Allows tutors to add, edit, or disable available times.

Manage Packages Page

Allows tutors to add, edit, or delete educational packages.

My Bookings Page

Displays all bookings received by the tutor.

My Reviews Page

Displays reviews submitted by students.

Admin Pages

Admin Dashboard

Displays system statistics such as number of users, tutors, bookings, payments, and complaints.

Pending Tutors Page

Allows admin to approve or reject tutor verification requests.

Manage Users Page

Allows admin to view system users.

Manage Bookings Page

Allows admin to view all bookings.

Manage Payments Page

Allows admin to view payments, commission, and tutor earnings.

Manage Reviews Page

Allows admin to view reviews.

Manage Complaints Page

Allows admin to view complaints and update complaint status.

Manage Subjects Page

Allows admin to create, update, and delete subjects.

6. Template Usage

A full front-end template is allowed and may be used in the project to improve the visual design and user experience.

The template can be customized using:

- HTML
- CSS
- JavaScript
- Bootstrap

The template should be modified to match the system idea, colors, pages, and user roles. All required pages should be included and connected to the system features

Conclusion

Top Students Tutoring Platform is a complete web-based system that connects students with verified top-performing student tutors. The platform provides an organized learning environment that includes tutor verification, subject search, profile management, booking, payment tracking, online meeting links, reviews, complaints, and admin management.

The system is designed using ASP.NET Core MVC, SQL Server, Entity Framework Core, Code First Approach, and Bootstrap. It supports multiple roles and provides a strong foundation for future enhancements such as real payment gateway integration, wallet system, coupons, advanced notifications, and smart tutor recommendations.